



Mohan K

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Professional Summary:

- ✓ IT professional with **12 years** of experience in the software development lifecycle, specializing in business requirements analysis, application design, development, testing, implementation, maintenance, and documentation. Successfully handled complex projects and delivered high-quality solutions.
- ✓ Big Data expert with a focus on designing and architecting Hadoop applications, making informed technology recommendations, processing large volumes of data, and developing robust, scalable architectures.
- ✓ Skilled in analytical abilities and gathering comprehensive business requirements, translating client needs into well-documented applications that align with business objectives.
- ✓ Proven ability to thrive in dynamic environments, delivering projects on time and within budget through strong organizational, time-management, and planning skills.
- ✓ Deep understanding of AWS services (Athena, EMR, EC2) for efficient Big Data Analytics processing, ensuring scalable, reliable, and cost-effective cloud solutions.
- ✓ Five years of experience in Azure technologies, proficient in Azure Data Factory, Databricks, Synapse Analytics, SQL Database, Cosmos DB, and Machine Learning for scalable and highly available data solutions.
- ✓ Extensive knowledge in data ingestion pipeline design, Hadoop architecture, data modeling, data mining, and advanced data processing techniques, enabling robust, optimized data solutions.
- ✓ Proficient in Hive, Sqoop, Scala, Python, SQL, and Pyspark, with experience in AWS and Azure services such as EMR, EC2, S3, Athena, Glue, Redshift, Lambda, Azure Data Factory, Databricks, Synapse Analytics, Big Query, Dataproc, Dataflow, Bigtable, Cloud Storage, Cloud SQL, Cloud Identity and Access Management (IAM)
- ✓ Familiar with Agile and Waterfall methodologies and sprint planning, ensuring effective project lifecycle management and stakeholder value delivery.
- ✓ Developed Spark-based applications for low-latency streaming data ingestion using Kafka and Pyspark, enhancing ability to architect and optimize real-time Big Data solutions.
- ✓ Committed to staying updated with the latest technology trends and advancements to bring current, effective solutions to every project, driving innovation and efficiency.

Technical Skills:

✓ **Big Data Technologies:**

- **Hadoop Ecosystem:** HDFS, Hive, HBase, Sqoop
- **Processing Frameworks:** Spark, PySpark
- **Streaming and Messaging:** Kafka
- **Programming Languages:** Scala, Python
- **Visualization:** Tableau 10
- **Scripting:** Unix Shell Script
- **AWS Services:** EMR, EC2, ECS, ECR, Lambda, S3, Athena, Glue, Redshift, Step Functions, SSM, SQS, SNS, SES, CodeCommit, EventBridge, CloudWatch, CloudFormation, DynamoDB, DMS
- **Azure Services:** Azure Data Factory, Databricks, Synapse Analytics, HDInsight, Cosmos DB, Azure DevOps, Azure SQL Database, Azure Blob Storage, Azure Kubernetes Service (AKS)
- **GCP Services:** Big Query, Dataproc, Dataflow, Bigtable, Cloud Storage, Cloud SQL, Cloud Identity and Access Management (IAM)
- **Databases:** Azure SQL Database, Cosmos DB, Azure Blob Storage
- **Big Data Processing:** Hive, Sqoop, PySpark
- **Source Control and CI/CD:** Bitbucket, GitHub, CI/CD Pipelines
- **Workflow Management:** Apache Airflow

✓ **Deep Learning:** CNN, RNN, LSTM, NLP

✓ **Machine Learning:** Random Forest, Decision Tree, Classifications, Regressions, Artificial Intelligence

✓ **SAP BO Tools:** SAP Business Objects, SAP UDT, SAP IDT, SAP HANA, Dashboards, Design

Employment History:

- ✓ Worked in **Infosys** technologies from Oct 2012 to July 2019
- ✓ Worked in **Optimum Info systems** from July 2019 to Nov 2019.
- ✓ Worked in **OVATO technologies** from Dec 2019 to Mar 2021
- ✓ Worked in **Galax E Solutions** from November 2021 to March 2023
- ✓ Working in **EPAM Systems Pvt Ltd** from May 2023 to March 2024.
- ✓ Working in **Nucleus Teq** from Oct 2024 to Till Date.

Education:

- ✓ B. Tech (E.C.E) from **JNT University**, Hyderabad.

Project Summary

Project #8: Slack Migration of Redshift Tables to Iceberg Tables, from AWS to Trino Hive Iceberg Tables

- ✓ **Client:** Salesforce
- ✓ **Role:** Senior Data Engineer
- ✓ **Duration:** Oct 2024 to Till Date
- ✓ **Environment:** Python, Airflow, AWS S3, AWS Redshift, Apache Iceberg, Trino, Hive, HQL (Jupyter Notebook), PySpark, Power BI

Project Description: The Slack Migration of Redshift Tables to Iceberg Tables project aimed to modernize the data architecture by migrating and transforming large datasets stored in Redshift on AWS to a more efficient and scalable environment using Trino Hive Iceberg tables. This migration involved automating the ETL processes for 40+ Redshift tables containing employee data, user stories, bug data, and security-related information, which were later loaded into Iceberg tables for optimized querying and analytics. The project automated the migration, transformation, and CDC (Change Data Capture) processes to ensure data consistency and integrity while providing data for Salesforce's internal consumption and Power BI dashboards.

Key Contributions:

- ✓ **ETL Automation Using Airflow:** Designed and implemented Airflow Directed Acyclic Graphs (DAGs) to orchestrate the migration pipeline from AWS Redshift to Trino Hive Iceberg tables. These DAGs automated the extraction, transformation, and loading (ETL) processes, ensuring that data from the legacy Redshift system was transferred and transformed efficiently into the new Iceberg tables. The DAGs also incorporated CDC functionality to manage incremental data loads.
- ✓ **Data Extraction and Transformation (HQL in Jupyter Notebook):** Extracted data from Redshift using Hive Query Language (HQL) queries executed in Jupyter Notebooks. Data was cleansed, transformed, and enriched within the notebook environment before being loaded into Iceberg tables. This transformation involved complex business logic to ensure that all employee-related data (including user stories, bug reports, and security data) was accurately mapped and processed for migration.

- ✓ **Data Migration to AWS S3 and Iceberg Tables:** Migrated data from AWS Redshift to AWS S3 buckets, converting the data into Iceberg table format. This migration leveraged the scalability and high performance of Iceberg tables, ensuring efficient data storage and optimized query performance. The migration process was fully automated to ensure the transition was smooth and error-free.
- ✓ **CDC Data Loads and PySpark Jobs:** Developed and executed PySpark jobs within Airflow to automate Change Data Capture (CDC) for ongoing data migration. This ensured that new or updated records in the source system were automatically captured and loaded into the Iceberg tables without manual intervention. The pipeline was set up to run weekly, maintaining up-to-date data in the final consumption table, `dim_employees`, which holds the cleansed, transformed, and integrated employee data.
- ✓ **Final Consumption Table and Power BI Dashboard:** Once the data was migrated and transformed, it was stored in the `dim_employees` table. From this table, a View was created to provide the necessary data for Salesforce's Power BI dashboard team. This enabled stakeholders to access key metrics and insights related to employees, bugs, user stories, and security data. The dashboard provided visual analytics to aid decision-making within the organization.
- ✓ **Static and Incremental Data Loading:** The project was split into two phases: first, the static tables were loaded, ensuring a clean migration of historical data. Then, dynamic tables were loaded with real-time CDC data updates. This two-phased approach allowed for a seamless transition from Redshift to Trino Hive Iceberg tables while maintaining data integrity and consistency.
- ✓ **Automated Data Pipeline:** The entire data pipeline was automated, enabling the system to automatically load incremental data weekly, ensuring that Salesforce's internal systems and Power BI dashboards were consistently up-to-date. The automated nature of the pipeline also reduced the potential for human error and increased efficiency in data processing.

Project #7: Americold - The Orion Project

- ✓ **Client:** Americold
- ✓ **Role:** Senior Data Engineer
- ✓ **Duration:** May 2023 – March 2024
- ✓ **Environment:** Python, Pyspark, SQL, AWS (ECS, ECR, S3, Athena, EC2-Redhat Linux, CloudFormation, Glue, Redshift, Lambda, Step Functions, SES, SQS, SNS, SSM, Secrets Manager, Event Bridge, Code Commit, Code Pipeline, CloudWatch)

Project Description: The Americold Data Lake Orion Project aimed to transform data management and analytics capabilities within Americold, a leading provider of

temperature-controlled warehousing and logistics services. This initiative enhanced data processing, storage, and analytics, resulting in improved business insights and operational efficiency.

Key Contributions:

- ✓ Designed and executed end-to-end data processing workflows in AWS, managing Bronze, Silver, Staging, and gold layers for Americold's warehouse management system.
- ✓ Managed data ingestion from on-premise applications (C# with Oracle and PostgreSQL databases) to AWS Data Lake, ensuring seamless data transfer and processing.
- ✓ Converted XML data to JSON, cleansing and categorizing it into eight different segments.
- ✓ Built Jetparser, a robust Python application on Red Hat Linux EC2 servers, for continuous 24/7 file processing with autoscaling and bash scripting.
- ✓ Designed and implemented ETL pipelines using AWS Glue, creating eight PySpark jobs for data transformation and loading into Redshift staging tables.
- ✓ Created additional glue jobs and step functions to consolidate data from staging to gold layers, supporting billing and integration with data visualization and data science teams.
- ✓ Established data quality checks and validation processes to maintain data integrity.
- ✓ Defined and enforced data governance protocols, ensuring compliance with data privacy regulations and maintaining high standards of data quality and security.
- ✓ Automated data processing and ETL tasks using AWS EventBridge schedulers and step functions, ensuring efficient and reliable data movement.
- ✓ Developed stored procedures and views to optimize data processing within AWS Glue jobs.
- ✓ Monitored and optimized the performance of the data lake, fine-tuning database queries and implementing auto-scaling mechanisms for efficient resource utilization.
- ✓ Collaborated with cross-functional teams to ensure the scalability and performance of the Americold Data Lake Orion.
- ✓ Conducted training sessions for team members on using the Orion Data Lake for data analysis.
- ✓ Documented data engineering processes, best practices, and architectural decisions to support seamless maintenance and future expansion.
- ✓ Led the initial migration of data from Oracle databases to S3 buckets, ensuring a smooth transition from on-premise systems to the AWS cloud environment.
- ✓ Designed ETL processes to ingest and model data from various sources into the data lake.
- ✓ Diagnosed and resolved data-related issues, ensuring data availability and implementing robust error-handling mechanisms in data pipelines.
- ✓ Ensured continuous monitoring and maintenance of data ingestion and processing

systems to support Americold's operational needs.

Project #6: Customer Analytics Platform (CAP)

- ✓ **Client:** Optum
- ✓ **Role:** Senior Big Data Engineer
- ✓ **Duration:** Nov 2021 - Mar 2023
- ✓ **Environment:** HDFS, Hive, Sqoop, Scala, Unix Shell Script, Kafka, Azure Data Factory, Azure Databricks, Azure Synapse Analytics, Azure HDInsight, Azure SQL Database, Azure Cosmos DB, Azure Blob Storage

Project Description: Optum, Inc. is an American healthcare services provider. The project involved building ETL pipelines in AWS and migrating data to support digital healthcare, product development, automation, analytics, and healthcare operations.

Key Contributions:

- ✓ Developed PySpark applications to convert JSON data into data frames, enabling detailed element-level comparison.
- ✓ Used PySpark and Spark-SQL for efficient data extraction, transformation, and aggregation from various file formats.
- ✓ Assessed MySQL database schema and structures to plan an effective migration strategy.
- ✓ Executed migration to Azure services (Azure SQL Database, Azure Database for PostgreSQL, Azure Cosmos DB), involving data type conversions and schema mapping.
- ✓ Migrated data from MySQL to Azure using direct connectivity, data export/import, bulk copy, and Azure Data Factory.
- ✓ Created tables, schemas, and indexes in Azure, utilizing SQL scripts and data loading tools.
- ✓ Validated migrated data against source MySQL databases to ensure integrity and accuracy.
- ✓ Implemented security measures in Azure, including access controls and encryption.
- ✓ Optimized Azure data environment performance with indexes, partitions, and caching.
- ✓ Established monitoring and alerting mechanisms for performance tracking and availability.
- ✓ Set up backup and disaster recovery strategies for data protection and continuity.
- ✓ Implemented PySpark applications and Spark-SQL for efficient data processing and testing.
- ✓ Designed and scheduled tasks using Apache Airflow, managing dependencies and workflows.

- ✓ Conducted data quality checks to ensure compliance with data privacy regulations.
- ✓ Created and maintained documentation of data engineering processes and best practices.

Project #5: Ovato Digital Innovations Platform (ODIP)

- ✓ **Client:** Ovato
- ✓ **Role:** Data Engineer
- ✓ **Duration:** Dec 2019 - Mar 2021
- ✓ **Environment:** HDFS, Hive, Sqoop, Scala, Unix Shell Script, Kafka, AWS (EMR, S3, Athena, Pyspark, Glue, Redshift, Lambda), GitHub, Airflow, Autosys, CloudWatch

Project Description: OVATO is Australasia's leading media, marketing, and printing company, offering advanced data and digital technology solutions. This project focused on enhancing print and distribution efficiency while providing innovative marketing strategies.

Key Contributions:

- ✓ Developed in Utilizing AWS Database Migration Service (DMS) to migrate and replicate data from Oracle database to AWS S3 Data Lake. Worked in configuring and managing DMS tasks, performing schema conversions, ensuring data consistency, and troubleshooting migration issues
- ✓ Analyzing and optimizing ETL data pipelines and queries for improved performance and scalability tuning query performance, partitioning data, optimizing storage formats, or leveraging AWS services like AWS Glue ETL, Amazon Redshift Spectrum.
- ✓ Converting the JSON file data from Source system into data frames by using Pyspark and Spark-SQL, so the records can be easily compared at data element level
- ✓ Organizing data in S3 buckets, implementing data partitioning and versioning strategies, optimizing data access patterns, and ensuring data security and compliance
- ✓ Developing Pyspark scripts to process and analyze large-scale datasets stored in AWS Amazon S3, Redshift for data cleaning, data transformation, feature engineering, implementing machine learning algorithms using Pyspark libraries, and optimizing performance for distributed computing.
- ✓ Implementing data quality checks, data profiling, and data governance processes to ensure data accuracy, consistency, and compliance. This may involve using AWS Glue Data Brew for data profiling, establishing data quality metrics, and implementing data lineage and metadata management
- ✓ Developed Pyspark applications and Spark-SQL for data extraction, transformation and aggregation from multiple file formats.

- ✓ Monitoring the performance, troubleshooting issues, identifying bottlenecks and health of data engineering processes and infrastructure components using AWS CloudWatch.
- ✓ Designing, implementing, and optimizing data warehousing solutions including schema design, data modeling, query optimization, managing table distributions and sort keys, and ensuring efficient data loading from S3 data lake to Amazon Redshift.

Project #4: Enterprise Data Management Platform

- ✓ **Client:** Standard Chartered Bank (SCB)
- ✓ **Role:** Big Data-AWS Engineer
- ✓ **Duration:** Jul 2019 - Nov 2019
- ✓ **Environment:** HDFS, Spark, Hive, HBase, Sqoop, Scala, Unix Shell Script, AWS (EMR, S3, Athena, Glue, Redshift, Pyspark), GitHub, Airflow, Lambda, CloudWatch

Project Description: Standard Chartered PLC is a British multinational banking and financial services company. The project involved designing and architecting data processing pipelines using AWS services to enhance data management and analytics.

Key Contributions:

- ✓ Collaborate with stakeholders to understand business requirements and translate them into scalable AWS-based big data solutions.
- ✓ Design and architect data processing pipelines using AWS services such as Amazon EMR, Amazon Redshift, Amazon Athena, and AWS Glue.
- ✓ Develop strategies for data ingestion, transformation, storage, and analytics on AWS.
- ✓ Implement data ingestion processes using tools like AWS Data Pipeline, AWS Glue, or custom ETL scripts.
- ✓ Create and maintain data transformation workflows to cleanse, enrich, and preprocess raw data for analytics.
- ✓ Optimize data pipelines for performance, reliability, and cost-efficiency.
- ✓ Choose appropriate AWS storage solutions based on data requirements, including Amazon S3, Amazon DynamoDB, Amazon RDS, and more.
- ✓ Manage data partitioning, indexing, and optimization techniques to enhance query performance.
- ✓ Set up data retention policies, access controls, and encryption mechanisms for data security and compliance.
- ✓ Develop and manage scalable distributed processing solutions using Amazon EMR, Apache Spark, or AWS Lambda.

- ✓ Create and optimize SQL queries for data analysis using services like Amazon Athena or Amazon Redshift.
- ✓ Implement real-time data processing and streaming analytics using AWS Kinesis, Apache Kafka, or AWS Lambda.
- ✓ Monitor and troubleshoot AWS resources to ensure optimal performance and reliability.
- ✓ Implement logging and monitoring solutions using AWS CloudWatch, AWS CloudTrail, and third-party tools.

Project #3: I-alert Developments (ALL)

- ✓ **Client:** Ashok Leyland Limited
- ✓ **Role:** Big Data Developer
- ✓ **Duration:** Mar 2018 - Jul 2019
- ✓ **Environment:** HDFS, Hive, Sqoop, Scala, Unix Shell Script, Kafka, Azure Data Factory, Azure Databricks, Azure Synapse Analytics, Azure HDInsight, Azure SQL Database, Azure Cosmos DB, Azure Blob Storage

Project Description: Ashok Leyland Limited (ALL) embarked on its telematics initiatives to offer differentiated services to its customers. The project aimed to provide value-added features and services to fleet management solutions.

Key Contributions:

- ✓ Collaborate with stakeholders to understand business requirements and design scalable big data solutions on the Azure cloud platform.
- ✓ Architect data processing pipelines using Azure Data Factory, Azure Databricks, and Azure HDInsight for efficient data movement and processing.
- ✓ Design data models and schemas using Azure Synapse Analytics (formerly SQL Data Warehouse) for optimized query performance.
- ✓ Develop data ingestion processes using Azure Data Factory, Azure Functions, and other Azure services for reliable and efficient data movement.
- ✓ Transform and cleanse data using Azure Databricks or Azure Data Factory, ensuring data quality and consistency.
- ✓
- ✓ Utilize Azure Databricks for distributed data processing and analytics, implementing Spark-based data transformations and machine learning algorithms.
- ✓ Develop and optimize queries using Azure Synapse Analytics and Azure Analysis Services for interactive and batch analytics.

- ✓ Implement solutions using Hadoop and Spark ecosystems in Azure HDInsight for processing and analyzing large-scale datasets.
- ✓ Utilize Azure Stream Analytics for real-time data processing and streaming analytics.
- ✓ Choose appropriate Azure storage solutions like Azure Data Lake Storage, Azure Blob Storage, and Azure SQL Database based on data requirements.
- ✓ Design and implement data partitioning, indexing, and optimization strategies for efficient query performance.

Project #2: Operational Risk Reporting and Data Analytics

- ✓ **Client:** Bank of America
- ✓ **Role:** Big Data Developer
- ✓ **Duration:** May 2015 - Feb 2018
- ✓ **Environment:** HDFS, HBase, Spark, Hive, Sqoop, SQL, Linux ,Big query, Bigtable, Dataflow, Dataproc and GCS.

Project Description: Bank of America is a multinational banking and financial services corporation. The project involved creating reports for the BI team and exporting data into HDFS and Hive.

Key Contributions:

- ✓ Supported code/design analysis, strategy development and project planning.
- ✓ Created reports for the BI team using Sqoop to export data into HDFS and Hive.
- ✓ Export and Import data into HDFS, HBase and Hive using Sqoop.
- ✓ Involved in creating Hive tables, loading with data and writing Hive queries.
- ✓ Load and transform large sets of structured and semi structured data.
- ✓ Support code/design analysis, strategy development and project planning.
- ✓ Used tools like Apache Hadoop's DistCp if your data is in HDFS, or write custom scripts to read from your source and write to Bigtable using the Bigtable client libraries.
- ✓ For large datasets, consider using Google Cloud Dataflow, which can read from HDFS and write to Bigtable.

Project #1: Gujarat Human Resource Management Services (GHRMS)

- ✓ **Client:** GHRMS Program, Govt. Of Gujarat
- ✓ **Role:** SAP BO Senior Developer
- ✓ **Duration:** Oct 2012 - Apr 2015
- ✓ **Environment:** Business Objects XI R3.1, SAP BW, SAP ECC, Windows XP

Project Description: Gujarat Human Resource Management System (GHRMS) combines systems and processes to manage employee data. The project involved creating reports and dashboards for easy management and analysis of HR data.

Key Contributions:

- ✓ Creating Folder and subfolders for different categories (Senior Management and junior management) of reports and which will help in providing access to the users.
- ✓ Created user Prompts, conditions, and filters for various reports to specify the reporting parameters as per business requirements.
- ✓ Created Hierarchies from providing drill down options for the end-user.
- ✓ Created complex reports and user objects like measure objects by using @aggregate aware function to create the summarized reports.
- ✓ Created Report Level Variables, Objects and Formula to create new columns based on the result set.
- ✓ Created complicated reports including sub-reports, graphical reports, formula base and well-formatted reports according user requirements
- ✓ Analyzing and enhancing the existing BO Reports as per new requirements.
- ✓ Developed Web Intelligence reports by using of combined Queries.